

Subject: Phase 1 - Wilkinson: Next Generation CHERI Tag Controller

Phase 1 Project Selection Status Report

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CRSID: ksw40

Director of Studies: Dr John Fawcett

Please complete 1, 2 and 3 below.

1. Please write 100 words on your current project ideas.

Tagged memory implementations that use a tag table require a hardware tag controller to fetch both the value from main memory and the tag information from the table. Several existing controllers use different algorithms/schemes to store tags in the table. As a core objective, I aim to develop a tag controller simulator that can be used to evaluate two existing algorithms for tag table storage. The simulator should be able to determine the memory access patterns of the simulated tag controller for different workloads, and how these would translate to DRAM accesses if employed in a given memory hierarchy. As an additional objective, I aim to develop an analysis engine that will monitor the access patterns and look for patterns that lend themselves to compression and optimisation opportunities, suggesting these to the user of the engine. As an additional objective, I aim to develop a novel algorithm for tag table storage. As an additional objective, I aim to implement my algorithm in hardware, using a hardware description language, and verify its correctness.

2. Please list names of potential project supervisors, indicating any interactions you have had with them, for example: not contacted, awaiting reply, in discussion, agreed to supervise.

Dr Jonathan Woodruff; agreed to supervise

3. Is there any chance that your project will involve any computing resources other than the Computing Service's MCS and software that is already installed there, for example: your own machine, machines in College, special peripherals, imported software packages, special hardware, network access, substantial extra disc space on the MCS.

If so indicate below what, and what it is needed for.

I might require access to the Computer Lab's shared filespace to obtain particular files for my project. The files are simulation traces that I could use in my project.